

Designed by Remus Teodorescu 2026

LECTURE 10.1

Ionic Diffusion

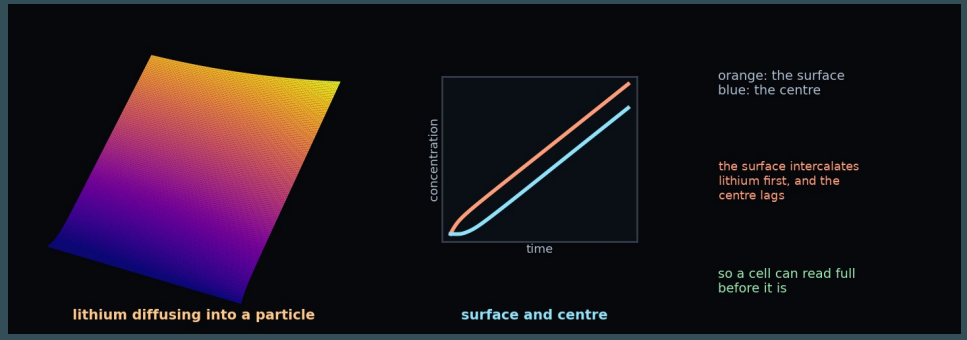
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Deep Learning for Engineering · 2026

PARTICLE · VOLTAGE · INVERSE



Lithium entering one particle: the surface intercalates lithium first and the centre lags, because lithium crosses only the surface and diffuses inward slowly. The lecture names the cell model's five state variables - lithium in particles and electrolyte, two potentials, temperature - then solves the particle, reads the state of charge, and recovers the diffusivity from the surface.

Learning Objectives

SECTIONS, IN ORDER

1. **The cell** — the battery, its state, current to voltage
2. **The particle** — diffusion, scaling, state of charge
3. **The complete model** — model levels, thermal coupling
4. **PINN and inverse** — the start built in, the diffusivity found

Incropera carries the transport and the diffusion, Liu the inverse method; Gu & Wang (2000) the coupling, Teodorescu et al. (2026) the BYD FC4680 parameters. The example is Ex_10.1's graphite particle; L10.2 takes the same physics to a solid oxide cell.

READ, AND WHAT EACH ANSWERS

Incropera ch. 14	how do lithium and charge move?
Incropera ch. 14	how does lithium reach the surface?
Incropera ch. 1	which model, and why does heat couple?
Liu ch. 4	what can readings identify?

Recap of L9.2 Gas Flow

1 – 2 · A NETWORK FOR THE CHANNEL

1 **How can a network learn the fraction of steam in the fuel channel, along it and across it, at any inlet speed?**

The network takes the position along the channel, the position across it and the inlet speed, and is trained until the species equation holds at points spread over the channel and the range of speeds. The rise of the mean along the channel is known from the hand calculation, so it is built into the form of the solution, and the network learns only the small profile across the channel, scaled by the wall flux so that it is of order one. *Liu ch. 2.2; Ex_09.2 section 5*

2 **Why does the network not keep the mean steam fraction of each cross-section right by itself, and how is it enforced?**

The hand calculation built into the solution gives the right mean, but the network can add a mean of its own, and a species equation satisfied only approximately at the training points does not prevent it. A loss term therefore holds the flow-weighted mean of the network's part across the channel at zero. With it the error falls from about a ten-thousandth to a few millionths of the steam fraction, more exact than any grid tested. *Incropera ch. 8; Ex_09.2 section 5*

3 – 4 · EVERY SPEED AND THE DESIGN

3 **What does a network gain by taking the inlet speed as an input, and how does it compare with finite differences?**

Trained once over a range of speeds, a few minutes offline, it gives the field at a new speed by one evaluation, where finite differences march the channel again. Compared online on the same device, no grid tested reaches its error, and against the finest grid it is a few times faster for the whole field and hundreds of times for one number. Its larger gain is not time but the speed as an input that can be differentiated. *Liu ch. 2.2; Ex_09.2 section 5*

4 **How can the inlet speed that leaves a required hydrogen fraction at the outlet be found with a network?**

Automatic differentiation through the trained network gives the derivative of the hydrogen fraction at the electrode at the outlet with respect to the inlet speed, and Newton's method uses it to find the speed for 25 % hydrogen: 0.3173 m/s against the exact 0.3174, in five steps with no further solve. It must start near the answer: from the hand calculation it converges, from one metre per second it diverges. A march answers the same question only by a sweep. *Liu ch. 2.2; Ex_09.2 section 6*

Lithium-Ion Batteries

Lithium-ion battery — a rechargeable cell whose lithium ions move through a liquid electrolyte between two electrodes that store them in their crystal lattice: graphite at the negative, an oxide or phosphate at the positive, which sets most of the energy, cost and lifetime.

MATERIALS AND PERFORMANCE

positive electrode	cell energy	cycles to 80 %	used in
NMC, NCA: nickel oxides	250 – 300 Wh/kg	1 000 – 2 000	long-range cars
LFP: iron phosphate	160 – 200 Wh/kg	3 000 – 6 000	cars, grid storage
LCO: cobalt oxide	180 – 230 Wh/kg	500 – 1 000	phones, laptops

Typical cell values, written from memory, not measured. The life depends on the depth of each cycle and the temperature as much as on the chemistry. LFP carries no nickel or cobalt and costs about 30 % less per kWh (IEA 2025).

THREE FORMATS



cylindrical — a jelly roll in a steel can: 18650, 21700, 4680

prismatic — rolled or stacked in an aluminium case

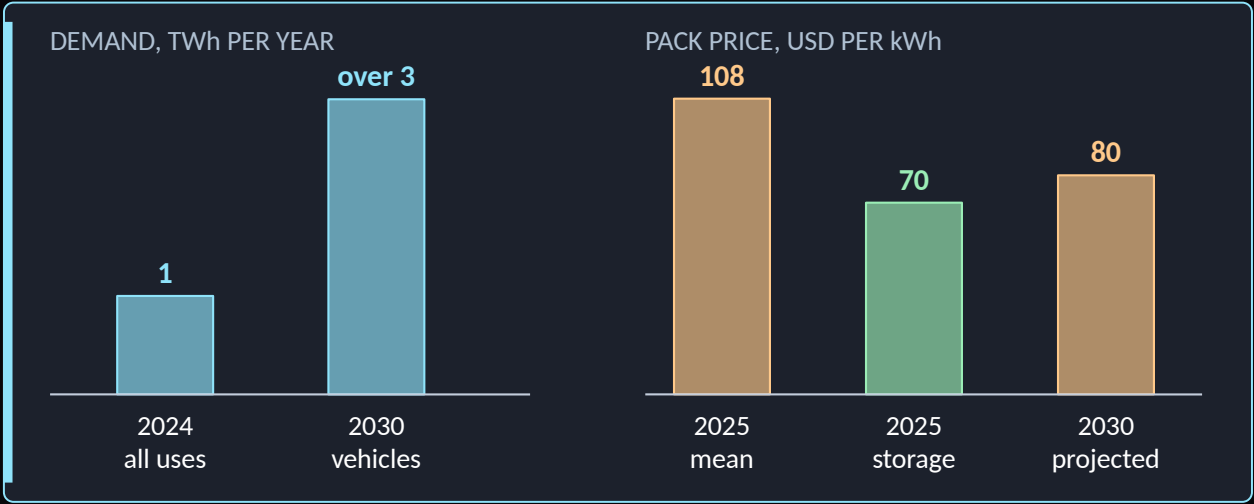
pouch — stacked sheets in a laminated foil

Incropera ch. 1; IEA, Global EV Outlook 2025

Demand Cost and Sustainability

Battery demand — the storage capacity of the cells made in a year. It reached about one terawatt-hour in 2024, over 85 % of it for electric cars, and the International Energy Agency expects electric vehicles alone to need over three by 2030, while the price per kilowatt-hour keeps falling.

DEMAND AND PRICE



2030 demand: electric vehicles only, the IEA's stated-policies scenario. 2030 price: BNEF's projection of 2023; the 2025 prices are its survey of December 2025.

APPLICATIONS AND SUSTAINABILITY

cars	over 85 % of vehicle battery demand
grid storage	the cheapest packs, USD 70/kWh
cobalt	two-thirds mined in DR Congo
graphite	over 90 % refined in China
EU, 2031	6 % recycled lithium, 16 % cobalt
EU, recovery	lithium 50 % by 2027, 80 % by 2031
EU, 2027	a digital passport per battery

Recycling cannot cut mining much for a decade: too few batteries have reached their end of life (IEA 2025).

IEA, Global EV Outlook 2025; BNEF 2025; EU 2023/1542

Inside the Cell

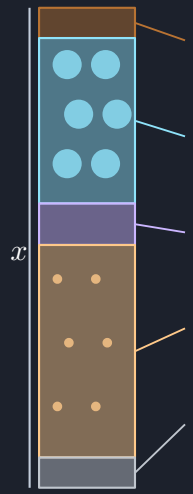
Jelly roll — two electrodes coated on metal foils, with a porous separator between them, wound into a cylinder and soaked in electrolyte. Lithium lives in two places: inside the electrode particles and dissolved in the electrolyte, so a model solves for five state variables on two length scales.

THE BYD FC4680 OF EX_10.1, UNROLLED



the can, end on
46 × 80 mm

unrolled: a strip 5.37 m long, 69 mm high



158 µm, to scale

copper foil, 10 µm
negative current collector

graphite, 54.5 µm
particles R = 4.80 µm, 31.9 % pores

separator, 13.7 µm
porous film, 47 % pores

$LiFePO_4$, 70.0 µm
particles R = 0.50 µm, 33.1 % pores

aluminium foil, 10 µm
positive current collector

pores: $LiPF_6$ in carbonates; foils coated on both sides

THE FIVE STATE VARIABLES

state variable	lives	equation
c_s in the particles	0 to R	parabolic
c_e in the electrolyte	0 to L	parabolic
φ_s in the solid	0 to L	elliptic
φ_e in the electrolyte	0 to L	elliptic
T, the temperature	the cell	parabolic

Every type in the table is one this course has solved: L8's heat equation and its Poisson problems, wearing electrochemical names. What is new is that the five supply terms to each other. Ex_10.1 solves the first row; the rest is this lecture's.

Diffusion

Diffusion — flow of a species from high to low concentration. Fick's first law gives the flux from the gradient; conservation turns it into the second law, for the concentration in time. A cell has it twice: in each particle along its radius, and in the electrolyte through the cell.

FICK'S TWO LAWS, IN THE SOLID AND THE ELECTROLYTE

$$\mathbf{J} = -D\,\nabla c \qquad \frac{\partial c}{\partial t} = -\nabla\cdot\mathbf{J} = \nabla\cdot(D\,\nabla c)$$

$\mathbf{J}$ the flux of lithium, c its concentration, D the diffusivity : first law, then the second by conservation

$$\frac{\partial c_s}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D_s \frac{\partial c_s}{\partial r}\right)$$

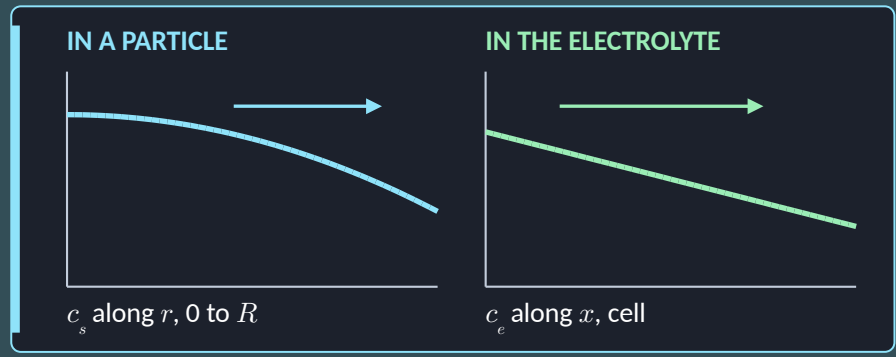
in the particle, along r

$$\varepsilon\,\frac{\partial c_e}{\partial t} = \frac{\partial}{\partial x}\left(D_e^{eff}\frac{\partial c_e}{\partial x}\right) + \frac{1-t_+}{F}\,j^{Li}$$

in the electrolyte, along x

c_e lithium in the electrolyte, ε the porosity, $D_e^{eff} = \varepsilon^{1.5} D_e$, t_+ the transference number
j^{Li} the reaction current per volume, F Faraday's constant; the particle's law is the same along r

TWO PLACES, THREE TIME SCALES



graphite particle	$R^2/D_s \approx 2\,400$ s
$LiFePO_4$ particle	$R^2/D_s \approx 4\,740$ s
electrolyte, negative	$L^2/D_e^{eff} \approx 93$ s

BYD FC4680 at 25 °C: the particles are the slow part, so Ex_10.1 solves the particle.

Charge Conservation

Charge conservation — one equation for the solid phase, for φ_s in both electrodes, and one for the liquid electrolyte, for φ_e in both electrodes and the separator. Neither has a time derivative: the potentials follow the current at once. The reaction moves charge from one phase to the other.

ONE EQUATION PER PHASE, AND THEIR SHARED SOURCE

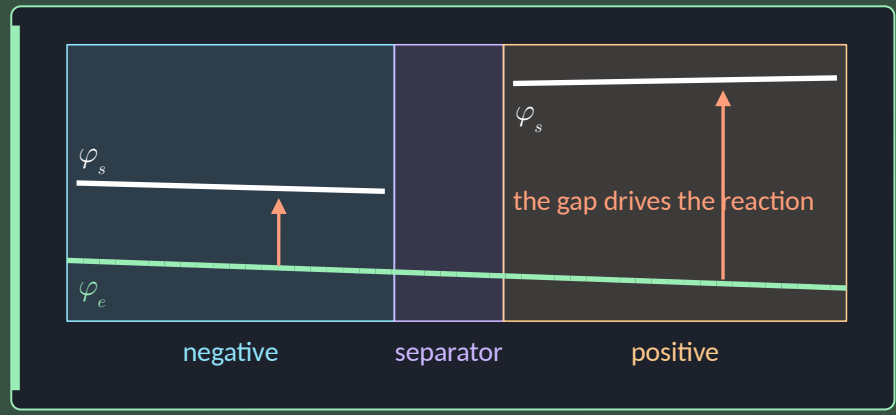
$$\nabla \cdot \left(\kappa^{eff} \nabla \phi_e\right) + \nabla \cdot \left(\kappa_D^{eff} \nabla \ln c_e\right) + j^{Li} = 0$$

$$\nabla \cdot \left(\sigma^{eff} \nabla \phi_s\right) - j^{Li} = 0$$

ϕ_e, ϕ_s potentials of the electrolyte and the solid, $\kappa^{eff}, \sigma^{eff}$ their conductivities, κ_D^{eff} the diffusional conductivity c_e the lithium in the electrolyte, j^{Li} the reaction current per volume, opposite signs

TALK · 2 MIN How would you explain, in your own words, what an equation with no time derivative says about its field? Then, in the drawing on the right, why is there no white line in the separator?

THE TWO POTENTIALS ACROSS THE CELL



the $\ln c_e$ term a gradient of c_e shifts φ_e

why the two ions move unequally

So the potentials need the concentrations: diffusion first, then charge.

Gu & Wang (2000)

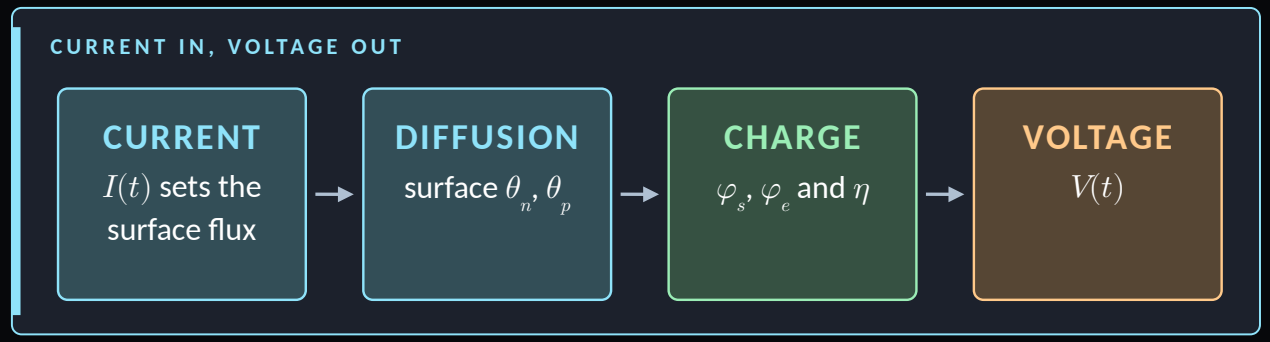
Voltage Response

Voltage response — the terminal voltage a cell gives for an applied current. The current sets the flux at the particle surfaces, diffusion gives the surface states of charge, the open-circuit curves turn them into voltages, and charge conservation subtracts the losses.

THE TERMINAL VOLTAGE, TERM BY TERM

$$V = U_p(\theta_p) - U_n(\theta_n) + \eta_p - \eta_n + \Delta\phi_e + \Delta\phi_s$$

U_p, U_n open-circuit potentials at the surface states of charge θ_p, θ_n (from diffusion)
η_p, η_n reaction overpotentials, $\Delta\phi_e, \Delta\phi_s$ drops in electrolyte and solid (from charge conservation)



WHAT EACH TERM COMES FROM

open circuit	the surface states of charge
reaction	η from the reaction kinetics
electrolyte	$\Delta\varphi_e$, with the $\ln c_e$ term
solid	$\Delta\varphi_s$, small in graphite

Ex_10.1 keeps the first pair: its film reads the voltage from the two surface states of charge. The losses belong to the fuller model levels.

Diffusion Inside a Particle

Solid diffusion — Fick's second law in a sphere: L8.2's heat equation in spherical coordinates, with symmetry at the centre and the reaction's flux at the surface. The reaction sees the surface concentration, not the average, so a particle can read as empty before it is.

THE EQUATION, AND ITS TWO CONDITIONS

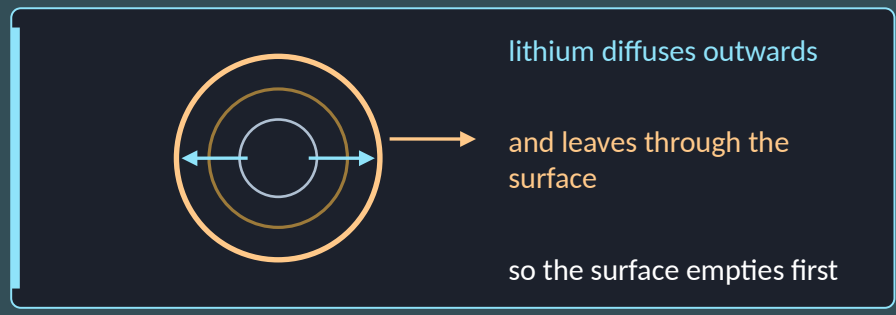
$$\frac{\partial c_s}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D_s \frac{\partial c_s}{\partial r}\right)$$

$$\left.\frac{\partial c_s}{\partial r}\right|_{r=0} = 0, \qquad -D_s \left.\frac{\partial c_s}{\partial r}\right|_{r=R} = j$$

c_s the lithium in the particle, r the radius, D_s the solid diffusivity, R the particle radius, j the surface flux

TALK · 2 MIN How would you explain, in your own words, why the reaction sees the surface and not the average? Then, in the drawing on the right, what happens to the gap between surface and centre when the current doubles?

EX_10.1'S GRAPHITE PARTICLE



	value
radius R	4.80 µm
diffusivity D_s	9.6×10^{-15} m²/s
diffusion time R^2/D_s	2400 s
discharge	0.9C: 13.87 A for an hour

The BYD FC4680, Teodorescu et al. (2026). The hour and the diffusion time are alike: the profile matters.

Scaling the Problem

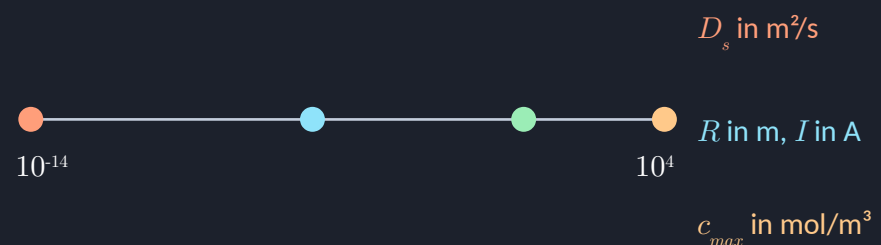
Scaling — measuring each quantity in a unit the problem itself supplies, so every number the network meets is of order one. In raw SI units these quantities span about twenty orders of magnitude, one residual term dwarfs the rest, and no loss weight makes up for that.

THE SCALED PROBLEM, AND ITS ONE CONSTANT

$$u_\tau = C\left(u_{rr} + \tfrac{2}{r}\,u_r\right), \quad C\,u_r = 1 \text{ at } r = 1, \quad C = \frac{D_s\,t_{end}}{R^2} = 1.50$$

u the lithium lost, over $c_{ref} = j\,t_{end}/R$, r over R, τ over the hour t_{end}, C the scaled diffusivity

THE RAW QUANTITIES, ONE LOG SCALE



THE SCALES, SUPPLIED BY THE PROBLEM

length	the radius R
time	the hour t_{end}
concentration	$c_{ref} = j\,t_{end}/R$
what is left	one number, C = 1.50

One free gift of these units: the mean of u over the sphere is exactly 3τ - what has left is the current times the time - whatever the diffusivity. The mass balance, and section 5 puts it in the loss.

State of Charge and Voltage

State of charge — how full the cell is. The bulk value is the mean concentration: a coulomb counter, exactly $1 - \tau$ here, needing no field. A gauge that has lost count must read the voltage, and the voltage sees only the surface - it is the solved field that converts the one into the other.

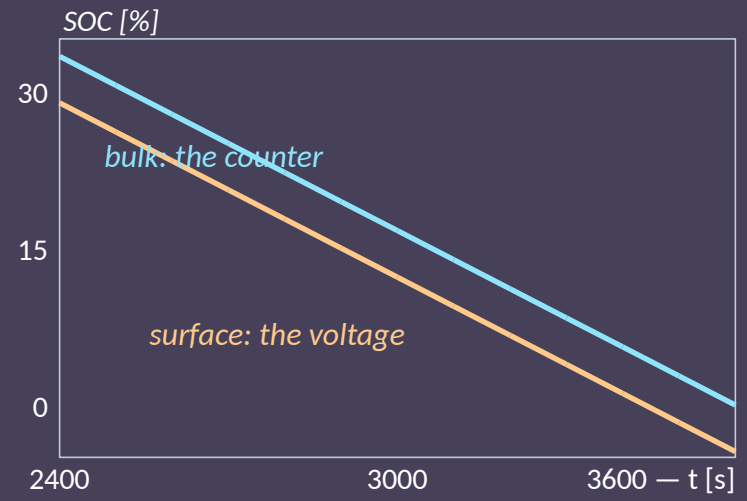
THE SURFACE SETS THE VOLTAGE

$$\theta = \frac{c_{se}}{c_{s,\max}} \qquad V = U_p(\theta_p) - U_n(\theta_n) + \eta_p - \eta_n$$

θ the surface state of charge, c_{se} the surface concentration, $c_{s,\max}$ the most a particle holds U_p, U_n the measured open-circuit curves, η_p, η_n the overpotentials (left out in Ex_10.1)

The open-circuit curves are measured, not derived: strongly nonlinear, with plateaus where two phases coexist. On LFP the plateau moves some 10 mV over most of the charge, so there the voltage says almost nothing about the surface - Ex_10.1's film shows it.

THE TWO GAUGES, OVER THE HOUR



Ex_10.1's exact field, the last 20 minutes: the surface reads about 4 points low and reaches zero almost three minutes early - the cell cuts off with charge in the bulk.

Three Model Levels

Model hierarchy — the three levels of a cell model: the single particle model (SPM) keeps one particle per electrode, the SPMe adds the electrolyte, the DFN keeps a particle at every point of the electrode. The level is chosen by the current, not by ambition.

PICK THE LEVEL BY THE CURRENT

model	solves	holds to
SPM	two particles, one current	about 1C
SPMe	adds c_e and φ_e along x	about 3C
DFN	a particle at every node of x	any rate

For a PINN the choice costs more than for a classical solver: each added state variable is another network, another residual, another loss term to balance. A matrix only grows.

WHAT THE EXERCISE KEEPS

solved for	one particle's $c_s(r,\, t)$
the current	shared evenly: j known
the voltage	algebra after the solve
left out	electrolyte, potentials, heat

In the single particle model only the diffusion is a differential equation. Charge conservation is trivial - identical particles share the current evenly - and the voltage is one line of algebra once the particle is solved. MP10.1A takes the full DFN on.

Thermal Coupling

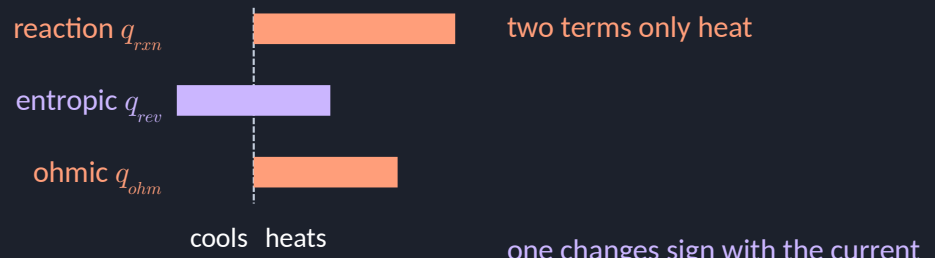
Thermal coupling — the loop between electrochemistry and temperature: the reaction, its entropy and the resistances make heat, and the temperature changes the diffusivities and rates that make it. A model with properties held at one temperature gets the trend backwards.

THE ENERGY EQUATION, AND ITS THREE SOURCES

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q, \qquad q = q_{rxn} + q_{rev} + q_{ohm}$$

ρc_p the heat capacity per volume, k the thermal conductivity, q_{rxn} reaction, q_{rev} entropic, q_{ohm} ohmic

THE THREE SOURCES



two terms only heat

one changes sign with the current

WHAT GU & WANG FOUND

coupled model	warming helps: transport improves
decoupled model	sees the heating, not the help
the result	the trend backwards
in a large cell	tens of degrees across it
for a PINN	one more residual term

The entropic coefficients were long unmeasured; Teodorescu et al. (2026) list them for the BYD FC4680.

A PINN for the Particle

Trial function with the start built in — a network wrapped so the initial condition holds whatever the weights do: τ times the network is zero at the start, as L8.2's pulse was. The residual is multiplied by r, so it has no $1/r$ in it and at the centre it reduces to the symmetry condition for free.

THE TRIAL FUNCTION, AND THE BALANCE IN THE LOSS

$$\hat{u}(r,\tau) = S\,\tau\,\mathcal{N}(r,\tau), \qquad \bar{\hat{u}} = 3\tau$$

$S = 3$ the size of the answer, $\tau\mathcal{N}$ zero at the start whatever the network does, $\bar{\hat{u}}$ its mean over the sphere

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4)      # r and tau in
u = 3.0 * tau * net(p)                          # the trial function: zero at the
start
g = grad(u, p)                                  # first derivatives by autograd
res = r * g[:,1:2] - C * (r * d2(u, p, 0) + 2 * g[:,0:1])   # the residual, times
r
mass = (u(q_pts) * q_w).sum(1) - 3 * tau_q      # the mass balance, held at 0, w
= 1000
```

THREE METHODS, ONE TRUTH

	FVM	PINN
surface at 3600 s	1683 mol/m³	1701 mol/m³
worst error	23 mol/m³	17 mol/m³
the hour, online	a few ms	under 1 ms
training, offline	none	about 3 min
agreed 30 mol/m³	40 shells	met

The exact series is the truth: surface 1684, centre 4164 mol/m³ at the end. The mass balance must be a loss term: a network a little late with the flux at the start loses lithium it never gets back. The surface points exclude $\tau = 0$, where the trial function has no gradient.

The Diffusivity From the Surface

Identifiability — whether the data can tell the parameter at all. The voltage reads the surface through the open-circuit curve, and the surface's lag behind the mean is what carries the diffusivity. On LFP's plateau that reading goes nearly blind.

ONE MORE TRAINABLE NUMBER

$$C \text{ trainable}, \qquad \mathcal{L} = \mathcal{L}_{PDE} + w_S\,\mathcal{L}_{flux} + w_M\,\mathcal{L}_{mass} + w_D\,\mathcal{L}_{data}$$

$\mathcal{L}_{data}$ the misfit to 37 surface readings, $w_S = 10$, $w_M = 1000$ hold the flux and the mass balance

THE INVERSE PROBLEM IN PYTORCH

```
logC = torch.nn.Parameter(torch.log(C0))       # the diffusivity, trained with the weights
C = torch.exp(logC)                            # positive by construction
L = res(C).pow(2).mean() + W_D * (u_surf - readings).pow(2).mean()   # physics + data
L.backward(); opt.step()                       # one step moves weights and C together
```

WHAT EX_10.1 RECOVERS

the data	37 surface readings
the noise	50 mol/m^3 on each
the start	half the true value
found	D_s 0.30 % low, in 1.5 min
not divided by C	divided, C runs away

Unlike Ex_08.2: divided by C, a uniform particle fits the readings and C grows without limit.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
the SPMe	lithium in the electrolyte	Incropera ch. 14	above about 1C
Butler–Volmer kinetics	the rate at the interface	PyBaMM docs	the overpotentials
the full model, DFN	a particle at every point	Marquis et al. (2019)	above 3C; MP10.1A
heat in two dimensions	the core hotter than the can	Incropera ch. 5	large cells and packs
ageing and lost lithium	slower diffusion, lost capacity	O'Regan et al. (2022)	the BMS; MP10.1B

WHY THEY ARE LEFT OUT

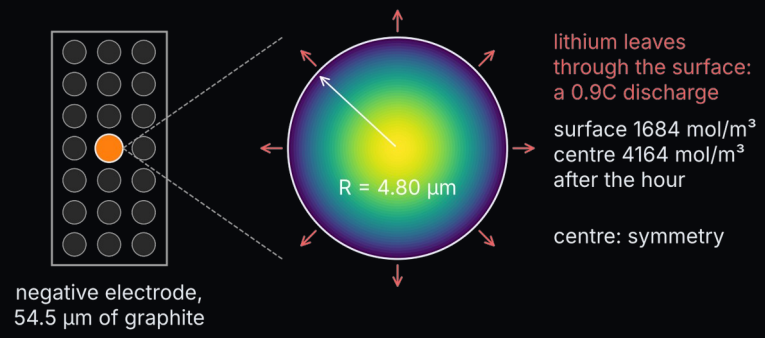
Ex_10.1 solves one particle with a known flux, scored on an exact series, so none of them is needed to do it. In the single particle model the current is shared evenly and the voltage is algebra; the rows return in the two mini projects.

EXERCISE

Exercise 10.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the particle, its data, its scales, the mass balance
- **4** — the exact series; finite volumes; the gauges and the film
- **5** — a PINN with the start built in; the three compared
- **6** — the diffusivity from 37 noisy surface readings
- **7 – 9** — the report; mini projects MP10.1A and MP10.1B



THE QUESTION EACH PART ANSWERS

1 – 3	which scales make it order one?
4	what does the surface hide?
5	can the PINN match the grid?
6	what do readings identify?

REPORT QUESTION, TO CONCLUDE

The grid solved the particle in milliseconds; the network matched it and then recovered the diffusivity from noisy readings in the same run. For which questions about a battery would you train a network, and for which solve a grid?

SUMMARY

Key Takeaways

THE CELL MODEL

what it is	five state variables, two scales
function	the voltage from the state
train, tune	pick the level by the current
vs one physics	each variable feeds the others
used for	what a BMS cannot measure

HEAT AND THE INVERSE

what it is	a particle PINN, start built in
function	the field, and D_s from readings
train, tune	scaled units; balance in the loss
vs finite volumes	equal accuracy, plus the inverse
used for	the parameter that ages most

Next — L10.2: the same transport and charge, in a ceramic cell at 800 °C.

Questions

1 – 2 · A NETWORK FOR THE PARTICLE

1 **How can a network learn the concentration of lithium inside the particle along the radius and in time?**

The network takes the radius and the time as inputs and returns the concentration. It is trained until the diffusion equation holds at points spread over the particle and the discharge, and the flux at the surface equals the one set by the applied current; no solved data is needed. The initial concentration and the symmetry about the centre are known exactly, so they are built into the form of the solution rather than learned. *Liu ch. 2; Ex_10.1 section 5*

2 **Why must the radius, the time and the concentration be scaled before the network is trained?**

In SI units the radius is a few micrometres, the diffusivity about 10^{-14} m^2/s and the concentration tens of thousands of moles per cubic metre: the terms of the loss differ by many orders of magnitude, and its gradient is dominated by the largest. Scaled by the particle radius, the discharge time and a reference concentration set by the current, every quantity is of order one, every term is weighted alike, and a single dimensionless constant is left to carry the physics. *Ex_10.1 section 3*

3 – 4 · CONSERVATION AND THE INVERSE

3 **Why does a network not conserve the amount of lithium in the particle by itself, and how is it enforced?**

A finite-volume solver conserves lithium by construction, because what leaves one shell enters the next. A network satisfies the diffusion equation only approximately at its training points, and small errors in the flux at the surface accumulate over the discharge into lithium gained or lost. The amount that has entered is known exactly from the applied current, so a loss term holds the mean concentration in the particle to that value, and the field stays physically possible. *Liu ch. 2; Ex_10.1 section 5*

4 **How can the solid-state diffusivity be estimated with a network from surface measurements, and when is it not possible?**

The diffusivity becomes a trainable parameter, optimised together with the network so that the concentration satisfies the diffusion equation and matches the surface measurements. It is identifiable because the lag of the surface concentration behind the mean depends on it. This works when the measurements change with the diffusivity; on the flat voltage plateau of LFP they barely change, and no method, network or classical fit, can identify it from the voltage alone. *Liu ch. 4; Ex_10.1 section 6*

NEXT

L10.2 — Solid Oxide Cells

The same transport and charge conservation in a ceramic cell at 800 °C, where L9.2's steam channel meets its electrode.

BEFORE THE NEXT SESSION

read	Incropera ch. 14
run	Ex_10.1, sections 1 to 6
bring	your diffusivity, and its error
think about	what a BMS can and cannot see

Play the film at 1C and at 5C before the session, and watch what the voltage does on the LFP plateau.